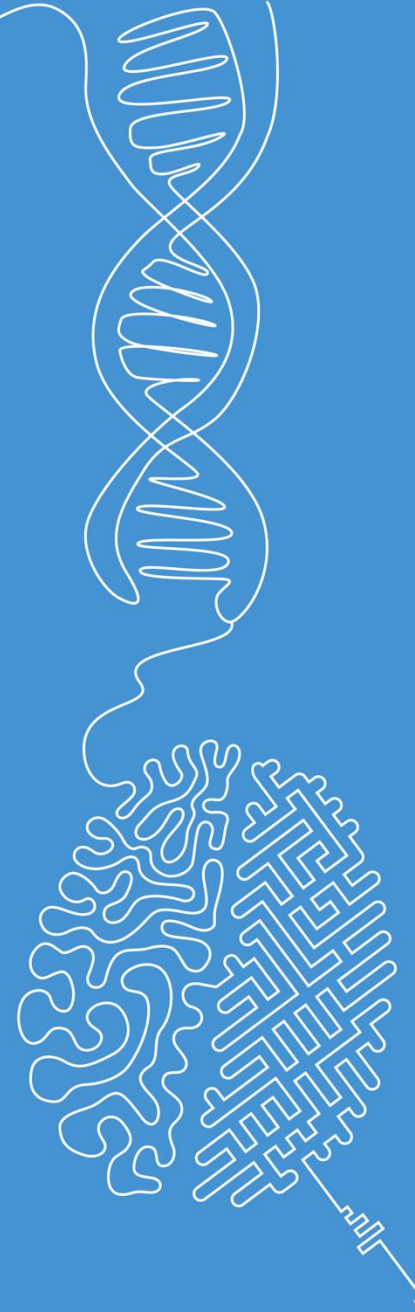


Fourier Series

Image and Signal Processing

Norman Juchler



Sinusoidal signals

Again, and again!

Sinusoidal signals (revisited)

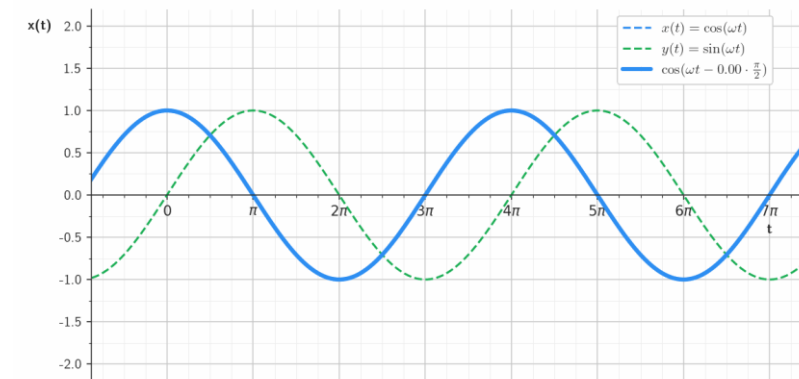
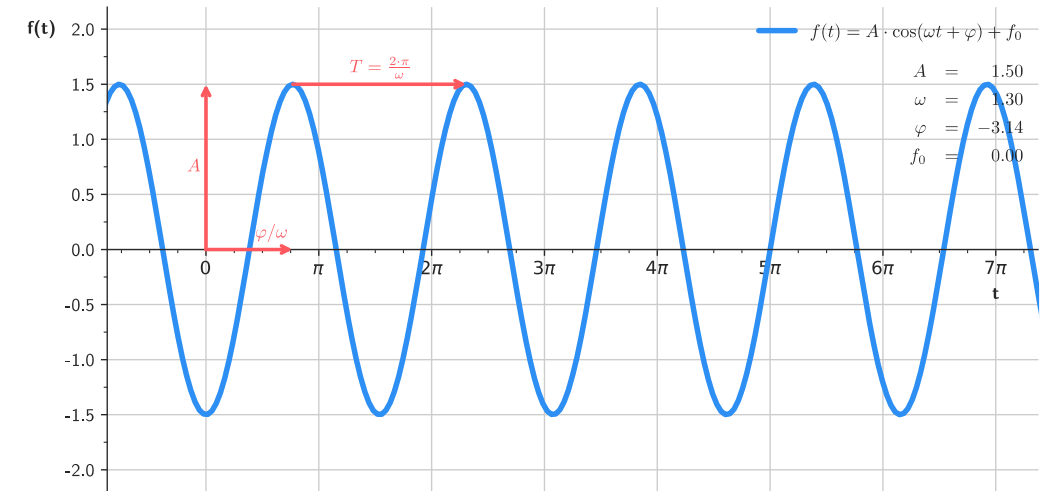
- The cosine takes this general form:

$$x(t) = A \cos(\omega t - \varphi) \quad \omega = \frac{2\pi}{T} = 2\pi f$$

ω : Angular frequency A : Amplitude
 f : Frequency φ : Phase
 T : Period

- This is equivalent* to

$$x(t) = a \cos(\omega t) + b \sin(\omega t) \quad \begin{aligned} a &= A \cos(\varphi) \\ b &= A \sin(\varphi) \end{aligned}$$

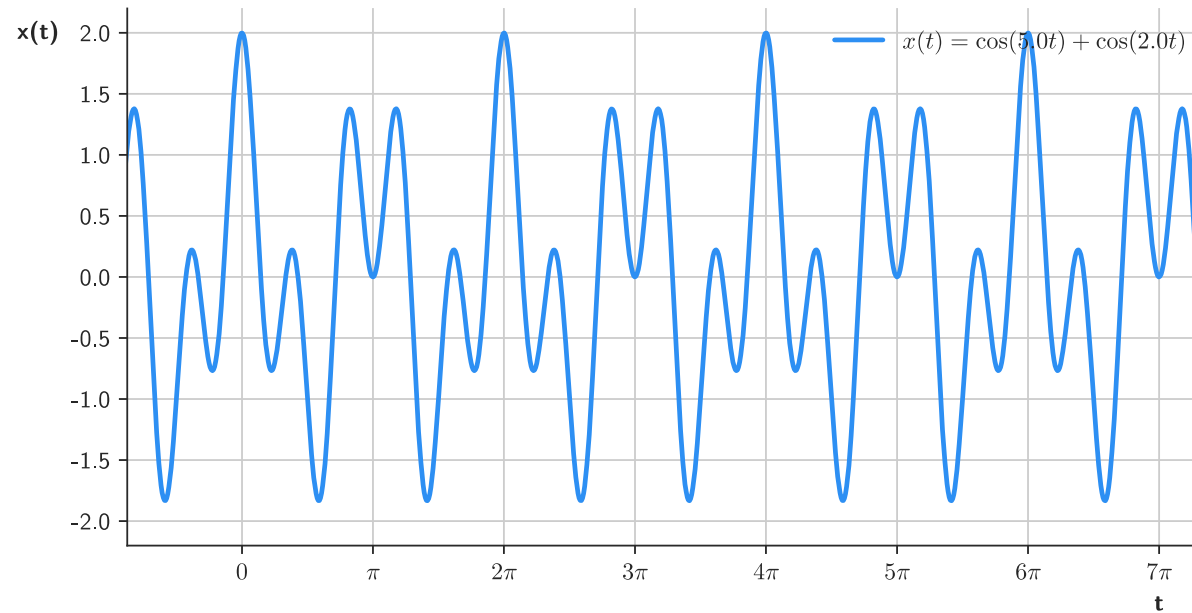


Sine and cosine are phase-shifted versions of the same waveform.

* The expression follows from this trigonometric formula:
 $\cos(\alpha - \beta) = \cos(\alpha) \cos(\beta) + \sin(\alpha) \sin(\beta)$

Interference: Two waves with different frequencies

- We can superimpose any number of sinusoidal signals
- **Example 1:** $x(t) = a_1 \cos(\omega_1 t) + a_2 \cos(\omega_2 t)$



Interference: N waves with different frequencies

- We can superimpose any number of sinusoidal signals
- **Example 1:** $x(t) = a_1 \cos(\omega_1 t) + a_2 \cos(\omega_2 t)$
- **Example 2:** More generally, we can superimpose any number of cosine signals:

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(\omega_k t + \varphi_k)$$

Here we have also added a vertical offset A_0

Interference: N waves with different frequencies

- We can superimpose any number of sinusoidal signals

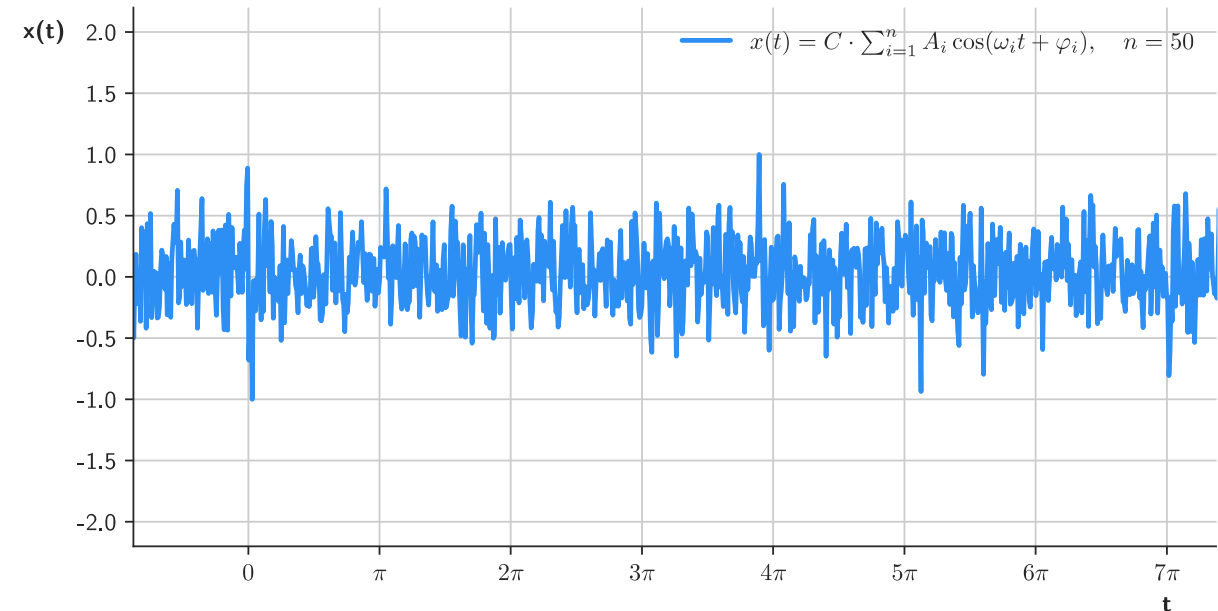
- **Example 3:**

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(\omega_k t + \varphi_k)$$

- Randomly sampled parameters

$$A_k, \omega_k, \varphi_k > 0$$

- Observation: For large N , the signal resembles a noisy signal



Note: The superposition has been scaled to fit within the range $[-1, 1]$ for better comparison.

Interference: N waves with different frequencies

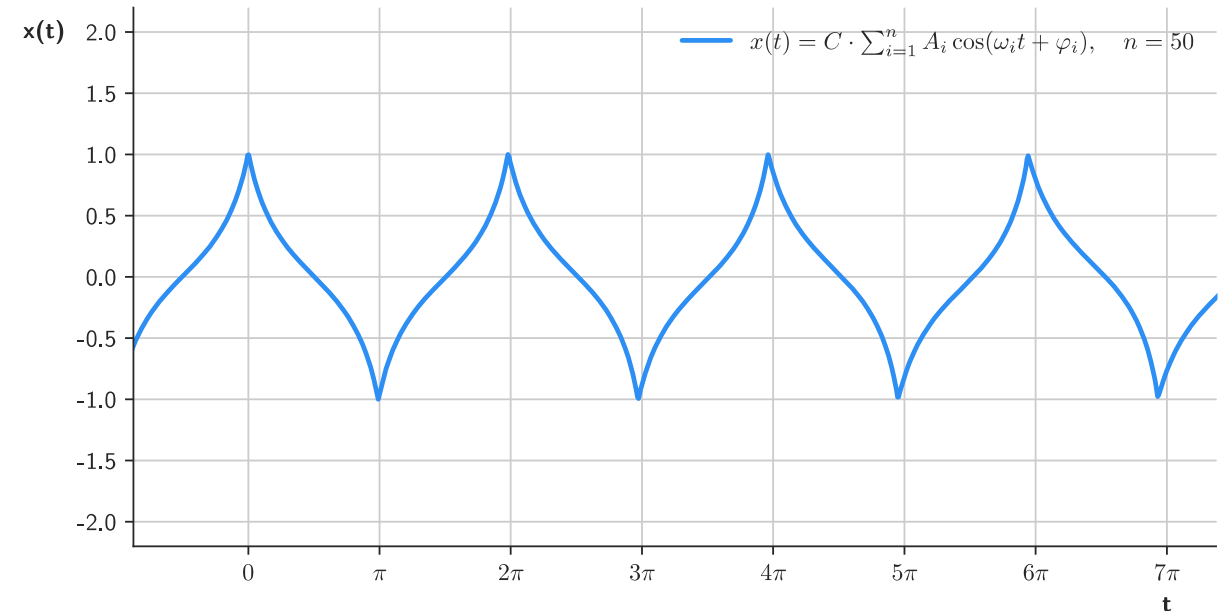
- We can superimpose any number of sinusoidal signals

- **Example 4:**

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(\omega_k t + \varphi_k)$$

- Here, the parameters are sampled according to the following rules:
$$A_k = \frac{c}{k^2}$$
$$\omega_k = \omega_0 + k \cdot \Delta\omega$$
$$\varphi_k = 0$$

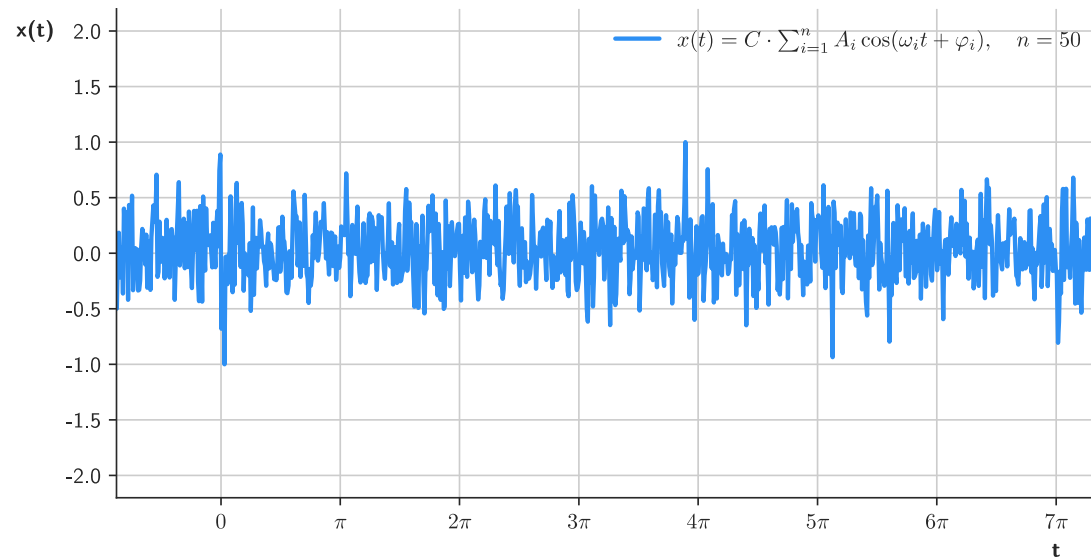
- Observation: As N increases, the sum of waves quickly converges to a periodic signal.



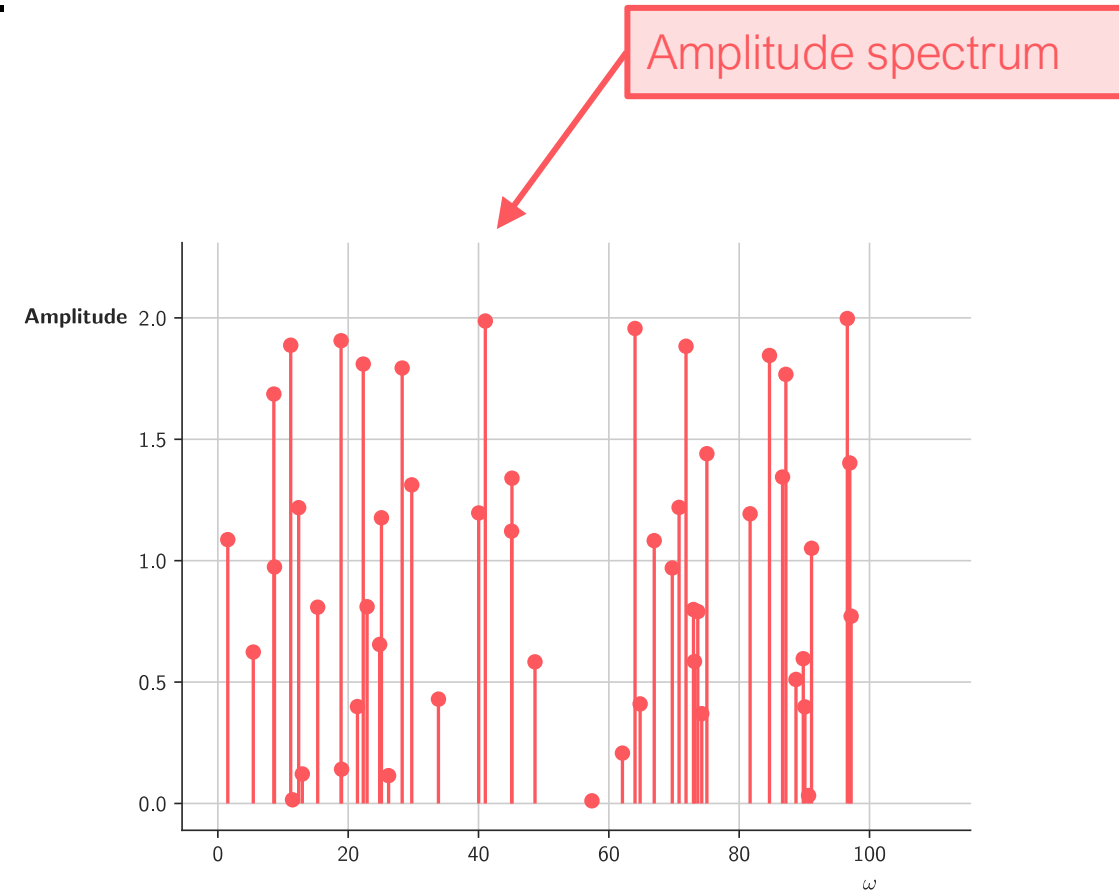
Note: The superposition has been scaled to fit within the range $[-1, 1]$ for better comparison.

The amplitude spectrum

- The amplitude spectrum represents the amplitudes of different frequency components in a signal.
- Plot: frequencies (f or ω) vs. amplitude
- Spectrum for **Example 2**:

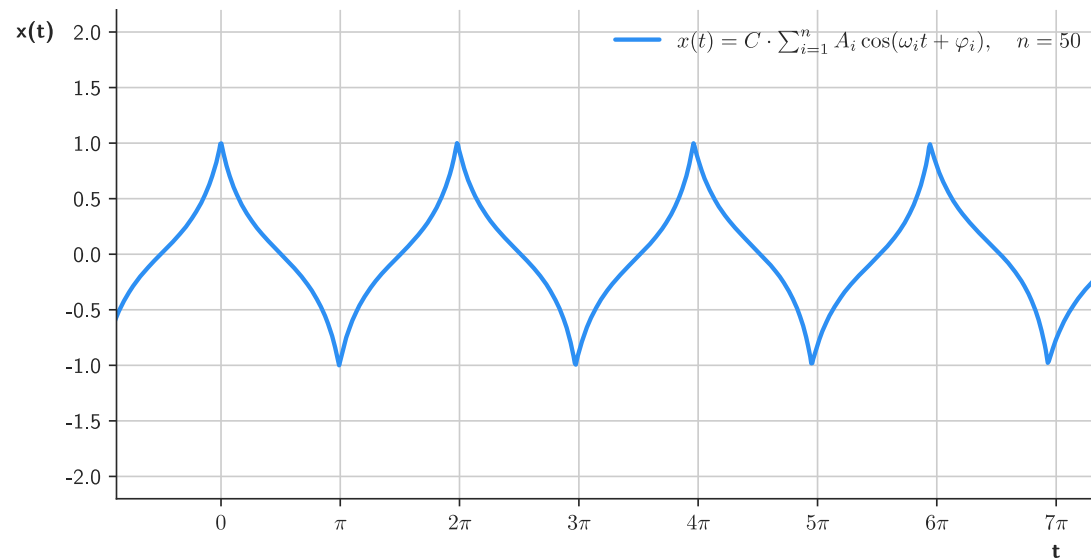


Note: The superposition has been scaled to fit within the range $[-1, 1]$ for better comparison.

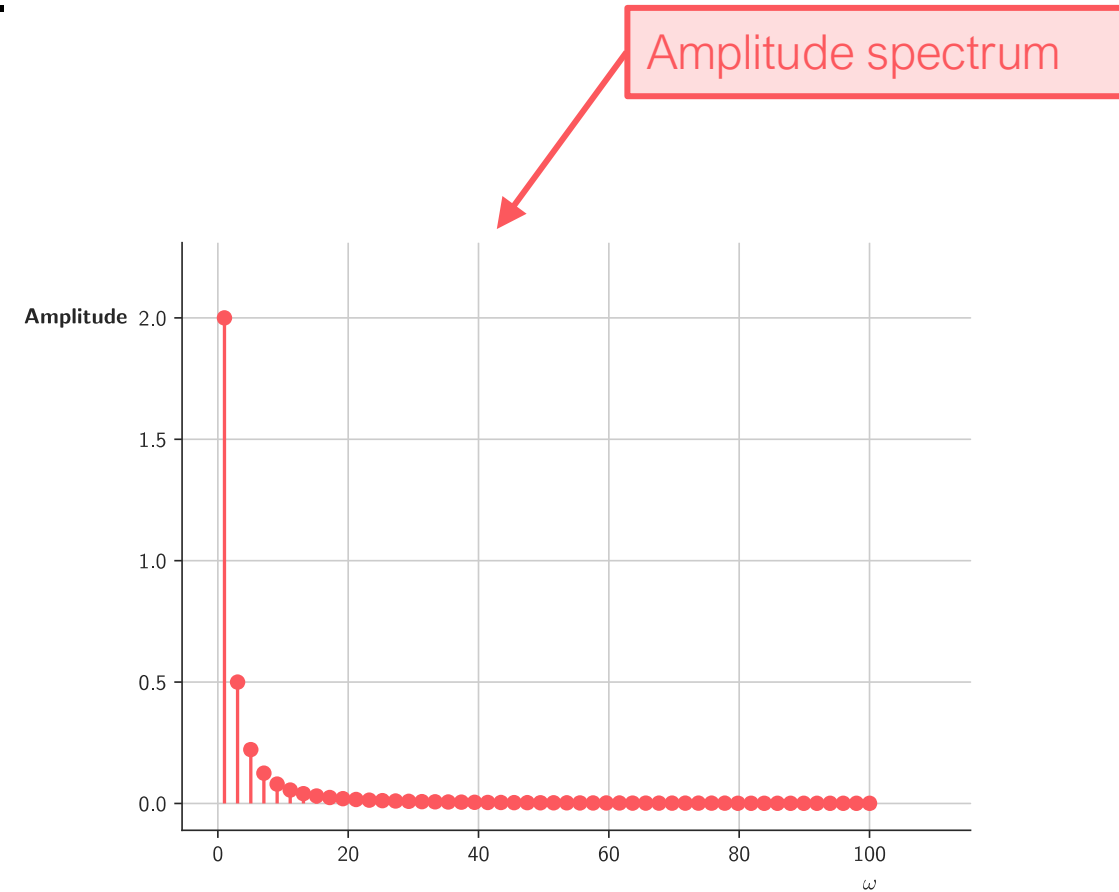


The amplitude spectrum

- The amplitude spectrum represents the amplitudes of different frequency components in a signal.
- Plot: frequencies (f or ω) vs. amplitude
- Spectrum for **Example 3**:



Note: The superposition has been scaled to fit within the range $[-1, 1]$ for better comparison.



Fourier series

Decomposing functions into harmonics

Fourier series

- The previous examples suggest that we can decompose signals into sinusoidal components.
- Fourier theory examines whether a (periodic) function or signal $x(t)$ can be expressed as a **series of trigonometric functions**:

$$x(t) = A_0 + \sum_{k=1}^{\infty} A_k \cos(\omega_k t - \varphi_k)$$

- How to determine the coefficients $A_k, k \geq 0$?
- To begin, we restrict ourselves to **periodic, continuous-time signals** $x(t)$ with period $T > 0$:

$$x(t + T) = x(t)$$



Joseph Fourier, 1768-1830, French mathematician and physicist. [See video](#)

Fourier series: Notation

- The following representations of the Fourier series are equivalent.

- **Fourier series in amplitude-phase form**

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos(\omega_k t - \varphi_k)$$

- **Fourier series in sine-cosine form**

$$x(t) = a_0 + \sum_{k=1}^N a_k \cos(\omega_k t) + b_k \sin(\omega_k t)$$

- **Fourier series, exponential form**

$$x(t) = \sum_{k=-N}^N c_k e^{i\omega_k t}$$

The last expression follows with some mathematical creativity using complex numbers.

Fourier series: Notation

- The coefficients of the different Fourier-series representations can be converted into each other.

- **Amplitude-phase form $\leftrightarrow$ sine-cosine form**

$$a_k = A_k \cos \varphi_k \quad \text{and} \quad b_k = A_k \sin \varphi_k$$

$$A_k = \sqrt{a_k^2 + b_k^2} \quad \text{and} \quad \varphi_k = \arctan \left(\frac{b_k}{a_k} \right)$$

- **Exponential form $\leftrightarrow$ sine-cosine form**

$$a_0 = a_0$$

$$a_k = c_k + c_{-k} \quad \text{for } k > 0$$

$$b_k = i(c_k - c_{-k}) \quad \text{for } k > 0$$

$$c_k = \begin{cases} a_0, & k = 0 \\ \frac{1}{2}(a_k - ib_k), & k > 0 \\ \frac{1}{2}(a_k + ib_k), & k < 0 \end{cases}$$

Excursion: Complex numbers

Definition: Complex numbers z are numbers of the form $z = a + ib$, $a, b \in \mathbb{R}$ with real part a and imaginary part b .

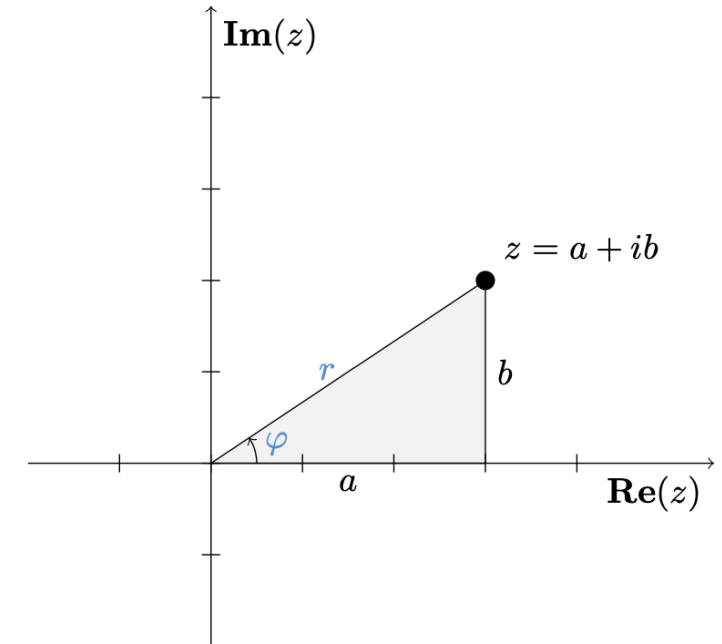
- They can be represented in the **complex plane**
- The fundamental laws of arithmetic apply: We can add, subtract, multiply, and divide complex numbers.
- A key identity is Euler's formula: $e^{i\varphi} = \cos \varphi + i \sin \varphi$
- Complex numbers can also be expressed in **polar form**

$$z = r(\cos \varphi + i \sin \varphi) \\ = re^{i\varphi}$$

Conversion
formulas:

$$r = \sqrt{a^2 + b^2} = |z| \\ \varphi = \arctan\left(\frac{b}{a}\right) = \arg(z)$$

$$a = r \cos(\varphi) \\ b = r \sin(\varphi)$$



Task: Check out the primer on complex numbers!

How to determine the coefficients?

- Frequency of the base functions increases with this pattern:

$$\omega_k = 2\pi f_k = \frac{2\pi k}{T}$$

- Fourier series in amplitude-phase form**

$$x(t) = A_0 + \sum_{k=1}^N A_k \cos \left(\frac{2\pi k}{T} t - \varphi_k \right)$$

- Fourier series in sine-cosine form**

$$x(t) = a_0 + \sum_{k=1}^N a_k \cos \left(\frac{2\pi k}{T} t \right) + b_k \sin \left(\frac{2\pi k}{T} t \right)$$

- Fourier series, exponential form**

$$x(t) = \sum_{k=-N}^N c_k e^{\frac{2\pi i k}{T} t}$$

The last expression follows with some mathematical creativity using complex numbers.

How to determine the coefficients?

- After one or two good digs into the mathematical bag of tricks, you will find the following expressions for the coefficients.

- **Sine-cosine form:**

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt$$

$$a_k = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos \left(2\pi \frac{k}{T} x \right) dt \quad \text{for } k \geq 1$$

$$b_k = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin \left(2\pi \frac{k}{T} x \right) dx, \quad \text{for } k \geq 1$$

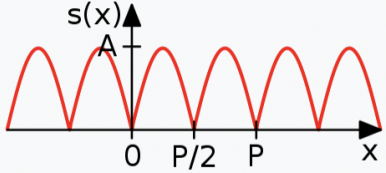
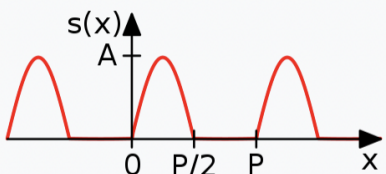
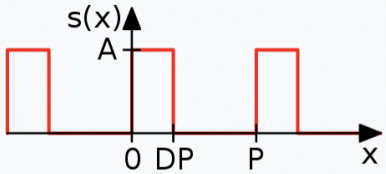
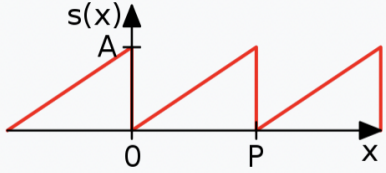
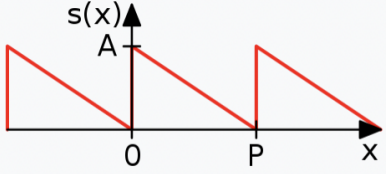
- **Exponential form:**

$$c_k = \frac{1}{T} \int_0^T e^{-i2\pi \frac{k}{T} t} x(t) dt, \quad \forall k \in \mathbb{Z}$$

Steps to apply the Fourier series

- Ensure the signal $x(t)$ is periodic and continuous-time.
- Determine the fundamental period T .
- Compute the Fourier coefficients:
 - by evaluating the above integrals
 - by using known results or symmetry properties
- Notes:
 - Each coefficient a_k, b_k, c_k corresponds to a discrete frequency $\omega_k = k\omega_0$.
 - The set of coefficients forms a discrete spectrum (amplitude spectrum).
 - Symmetry simplifies computation: Even signals $\rightarrow$ only cosine terms. Odd signals $\rightarrow$ only
- Explore
 - Examples in this week's Jupyter notebook
 - Text on symmetry properties in Fourier series integrals.

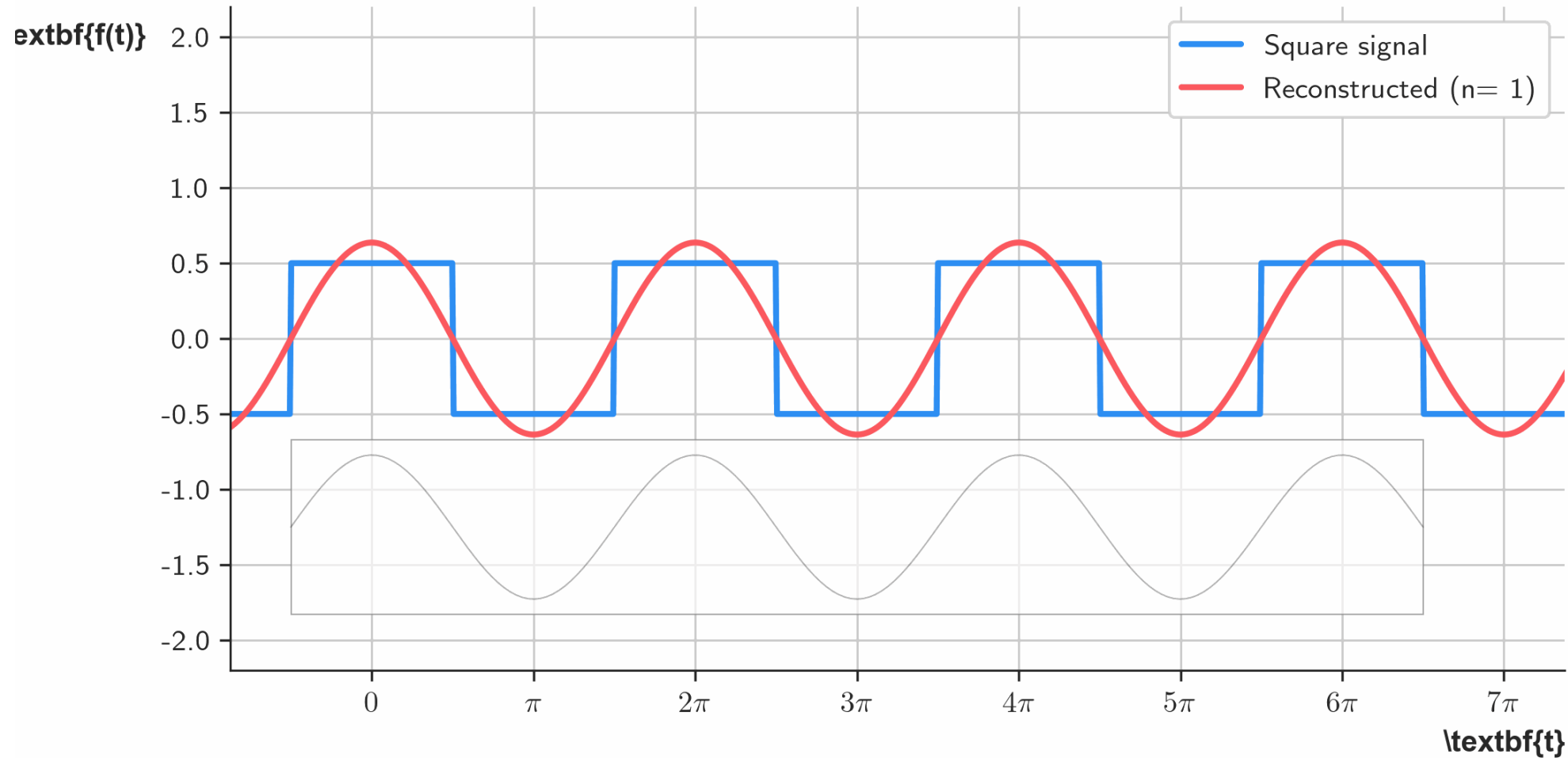
Fourier series: Extract of a formulary

$s(x) = A \left \sin\left(\frac{2\pi}{P}x\right) \right \quad \text{for } 0 \leq x < P$		$A_0 = \frac{2A}{\pi}$ $A_n = \begin{cases} \frac{-4A}{\pi} \frac{1}{n^2-1} & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$ $B_n = 0$
$s(x) = \begin{cases} A \sin\left(\frac{2\pi}{P}x\right) & \text{for } 0 \leq x < P/2 \\ 0 & \text{for } P/2 \leq x < P \end{cases}$		$A_0 = \frac{A}{\pi}$ $A_n = \begin{cases} \frac{-2A}{\pi} \frac{1}{n^2-1} & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$ $B_n = \begin{cases} \frac{A}{2} & n = 1 \\ 0 & n > 1 \end{cases}$
$s(x) = \begin{cases} A & \text{for } 0 \leq x < D \cdot P \\ 0 & \text{for } D \cdot P \leq x < P \end{cases}$		$A_0 = AD$ $A_n = \frac{A}{n\pi} \sin(2\pi nD)$ $B_n = \frac{2A}{n\pi} (\sin(\pi nD))^2$
$s(x) = \frac{Ax}{P} \quad \text{for } 0 \leq x < P$		$A_0 = \frac{A}{2}$ $A_n = 0$ $B_n = \frac{-A}{n\pi}$
$s(x) = A - \frac{Ax}{P} \quad \text{for } 0 \leq x < P$		$A_0 = \frac{A}{2}$ $A_n = 0$ $B_n = \frac{A}{n\pi}$

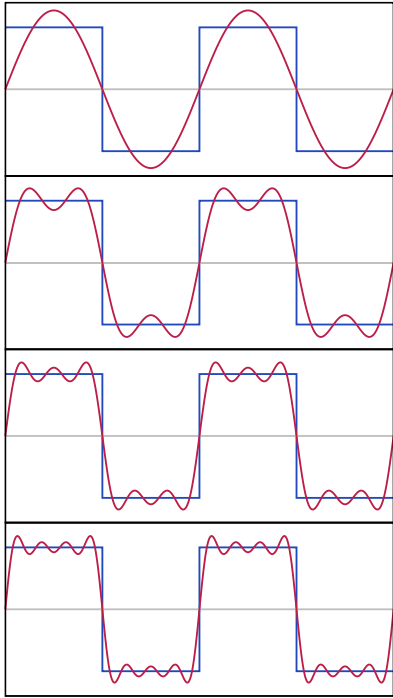
Fourier series

Examples, applications, outlook

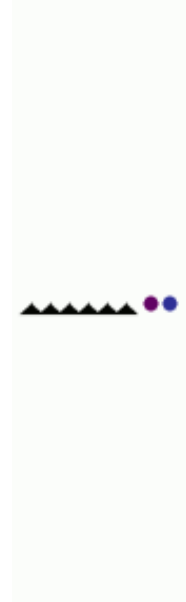
Example: Pulse function



Example: Pulse function



The first four partial sums of the Fourier series for a square wave. As more harmonics are added, the partial sums converge to the square wave. [Source](#)



[Animation](#) of the sixth partial sum (that is $N=6$) of the Fourier series for the square wave. The perfect square wave is represented by the blue dot on the right, while the approximated signal is shown by the purple dot. [Source](#)



Illustrative [animation](#) of the relationship between the Fourier series approximation (with $N=6$) of a square signal and its amplitude spectrum. [Source](#)

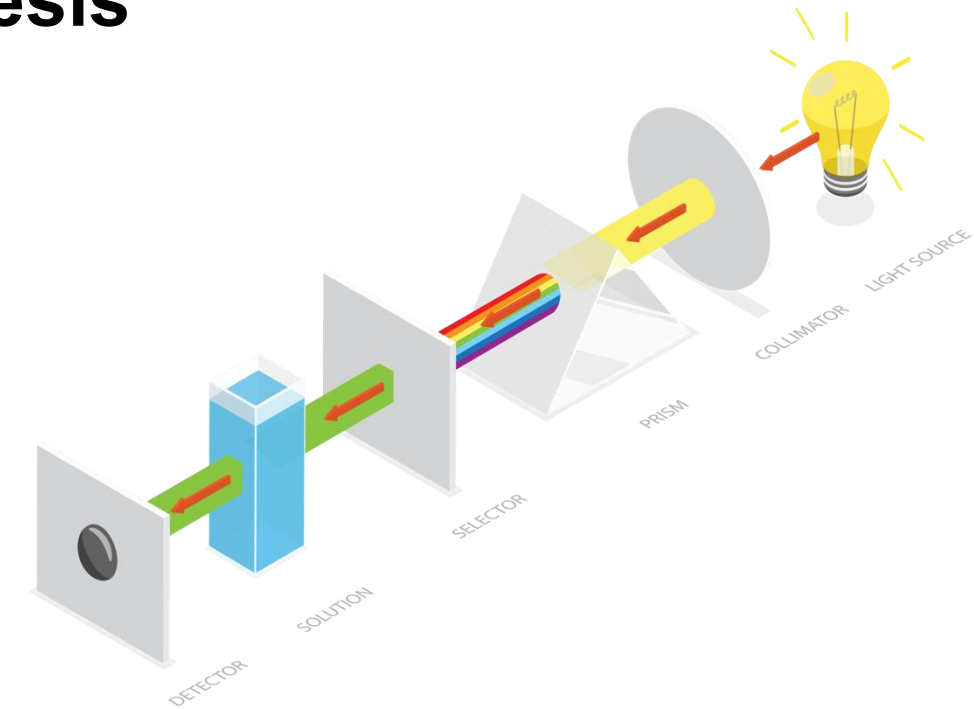
Application: Spectrum analysis vs. synthesis

■ Analysis:

- Time domain $\rightarrow$ Frequency domain
- Break down a signal into its individual frequency components to study its spectral content.
- Example: **Spectroscopic methods** are used to identify or describe chemical compounds by analyzing their absorption or emission spectra.

■ Synthesis:

- Frequency domain $\rightarrow$ Time domain
- Create new signals by combining individual frequency components or modifying existing ones.
- Example: **Audio synthesis** (sound design, speech synthesis)



Example for spectral analysis: Spectrography. The method permits to analyze molecules resolved in the test solution.

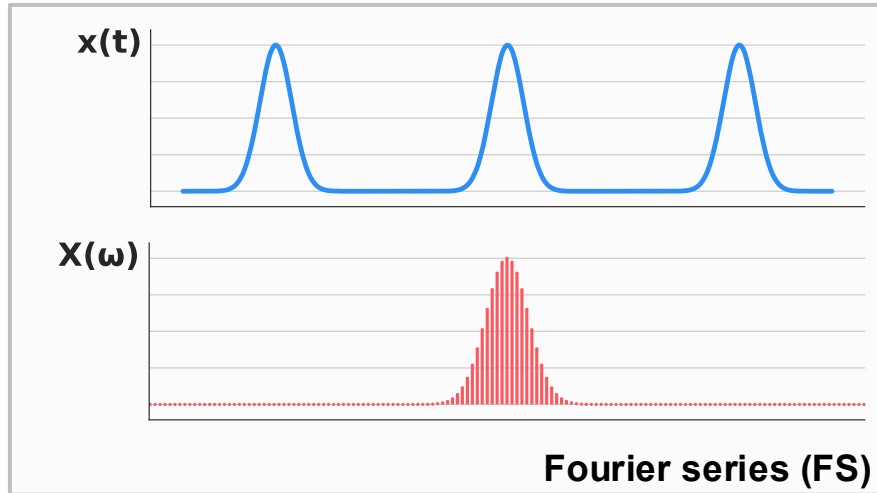
Outlook: From Fourier series to modern signal analysis

- Generalization to non-periodic signals $x(t)$:
Fourier transformation (FT) for general continuous-time signals
- Key properties of Fourier transformations:
 - Linearity
 - Time and frequency shifting
 - Convolution and the convolution theorem
- Discrete-time perspective:
 - Discrete-time Fourier transformation (DTFT)
 - Sampling and its implications on Fourier analysis
- Computational perspective:
 - Brief introduction to the FFT

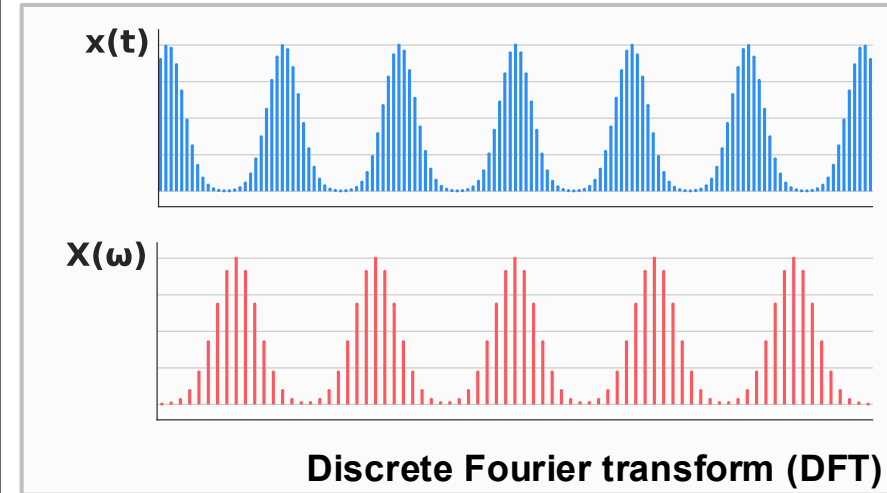
Outlook: Fourier's landscape

Periodic

Time-continuous



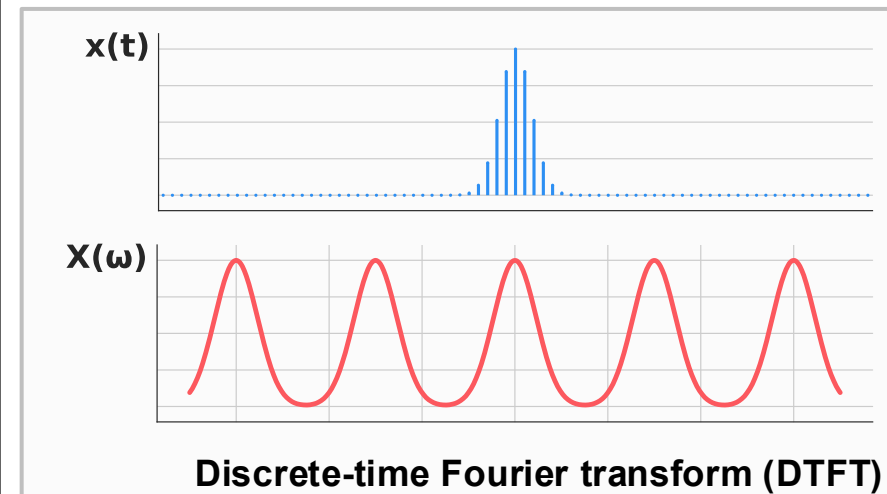
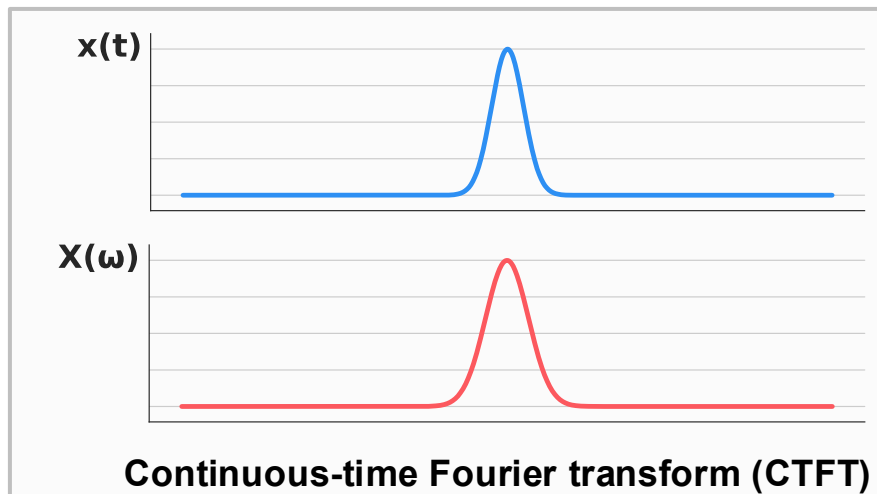
Time-discrete



Related:

- Fast Fourier transform
- Discrete cosine transform
- Discrete sine transform

Aperiodic



Generalization:
Laplace
Transform

Generalization:
z - Transform